

Uncertainty Signal Pilot: Perception and Reasoning Entropy

FERMAT Handwritten Math, Qwen2.5-VL-3B

August 2, 2026

Abstract

This pilot tests whether two entropy-based uncertainty signals – *perception entropy* (instability in transcribing a handwritten math answer) and *reasoning entropy* (instability in grading whether an answer contains an error) – carry real signal about model correctness, on 100 items from the FERMAT dataset with Qwen2.5-VL-3B. Perception entropy required two failed approaches before a working one was found; the final version achieves AUROC = 0.79 with clearly visible separation between correct and incorrect items. Reasoning entropy’s digit-only pipeline was clean from the start but the signal is weak (AUROC = 0.62 at $K = 5$); resampling to $K = 15$ confirmed the signal is real (the exact tie between correct/incorrect medians at $K = 5$ breaks) but only modestly stronger (AUROC = 0.665). A follow-up experiment clustering the model’s own **Reasoning** explanation text – rather than only its parsed **Error** digit – found a substantially stronger signal at $K = 5$ (AUROC = 0.702, without any additional sampling), traced to a concrete cause: in 12% of samples the model’s final digit contradicts its own just-written reasoning, and the digit is the noisier of the two. A GPU-accelerated confirmation at $K = 15$ reproduced the $K = 5$ number exactly but found the technique does *not* generalize: AUROC collapsed to 0.520 (worse than digit-only), traced to transitive cascade over-merging in the clustering algorithm that gets worse as the number of pairwise comparisons grows. The $K = 5$ result stands on its own; the technique needs a structural fix before further use at higher K .

1 Goal

Given a vision-language model transcribing and grading handwritten math answers, does the entropy of its own repeated outputs – how much it disagrees with itself across independent samples – predict when it is actually wrong? This is a fast, scoped-down pilot to check whether either signal exists at all, not the full study: no bootstrap confidence intervals, no baseline suite, no conformal calibration, no human double grading, no full-dataset run.

2 Setup

- **Data:** 100 items sampled (seed 42) from the FERMAT dataset.
- **Model:** Qwen2.5-VL-3B-Instruct.
- **Perception:** the model transcribes a handwritten Question–Answer pair; $K = 5$ samples at temperature 0.7, plus 2 samples at temperature 0 as a determinism sanity anchor.
- **Reasoning:** the model grades whether the handwritten Answer contains an error (binary 0/1); same $K = 5 + 2$ temperature-0 sampling scheme.

- **Entropy:** Shannon entropy in nats over the cluster distribution of the K sampled outputs for one item, $H = -\sum_i p_i \ln p_i$.
- **Correctness labels:** transcription is compared against the ground-truth answer; grading is compared against the ground-truth error label.

3 Perception Entropy: Two Failures, One Fix

3.1 Attempt 1: exact-string clustering (failed)

The transcription prompt asks for the *full* handwritten derivation, not just a final answer. Clustering by exact string match after light normalization treated any paraphrasing of that derivation as a different answer.

Items pinned at maximum entropy ($\ln 5 \approx 1.609$) 83/100

83% of items showed the maximum possible disagreement regardless of whether the model actually agreed with itself on the substance of the answer – the metric carried essentially no information.

3.2 Attempt 2: naive semantic clustering (failed)

Reasoning that exact-match was too strict, the next attempt clustered by semantic similarity instead: bidirectional NLI entailment and, separately, embedding cosine similarity, both applied to the whole derivation.

	Canonical (Attempt 1)	NLI	Embedding
Mean entropy	1.535	0.137	0.185
Items at max entropy	83/100	0/100	0/100

This looked like a fix – entropy dropped sharply – but the correctness rate that came with it (majority-vote transcription matched against ground truth via the same NLI entailment check) was $92/97 = 95\%$, implausibly high next to the model’s other performance figures. Inspection showed the cause: with premise and hypothesis sharing nearly identical scaffolding (the same LaTeX structure, the same boilerplate phrasing), the NLI model’s well-documented lexical-overlap bias pushed it toward “entailment” regardless of whether a buried number or coefficient actually differed. The fix for Attempt 1’s problem introduced a new, opposite failure mode: over-merging.

3.3 Attempt 3: final-answer extraction (worked)

The actual defect in both prior attempts was comparing the *wrong span of text*: the full derivation, when only the final answer matters for a correctness judgment. Attempt 3 extracts just the final answer – the last multiple-choice option statement, a `\boxed{}` result, the last display-math block, or (rarest) the last line – before clustering with the same syntactic/canonicalization approach as Attempt 1.

Metric	Value
AUROC (perception entropy $\rightarrow$ transcription error)	0.788
Items at max entropy	14/100
Transcription correct	42/100
Median entropy, correct group ($n = 42$)	0.500
Median entropy, incorrect group ($n = 58$)	1.332
Mean entropy, correct group	0.583
Mean entropy, incorrect group	1.099

This is the strongest result in the pilot: real, visible separation between correct and incorrect items (Figure 1, left panel).

4 Reasoning Entropy: Clean Pipeline, Weak but Real Signal

Unlike perception, the reasoning/grading pipeline required no comparable fix. It clusters a single parsed binary digit (error present/absent) rather than free text, so it never had the “wrong span” problem. The pipeline was independently re-verified by recomputing correctness from scratch for all 100 items against the stored results: zero mismatches.

4.1 $K = 5$ baseline

Metric	Value
AUROC (reasoning entropy $\rightarrow$ grading error)	0.618
Grading accuracy	75%
Median entropy, correct group ($n = 75$)	0.5004
Median entropy, incorrect group ($n = 25$)	0.5004

The medians are *exactly* tied. With only 5 samples of a binary outcome, there are just 3 realistic entropy values (unanimous, a 4–1 split, a 3–2 split), which structurally limits how well two groups can be told apart by this statistic.

4.2 $K = 15$ follow-up: is the weak signal a real ceiling, or under-sampling?

To test whether $K = 5$ was simply too coarse to reveal a real effect, grading was resampled to $K = 15$ for the same 100 items (reusing the original 5 samples and drawing 10 more), keeping perception untouched.

Metric	$K = 5$	$K = 15$
AUROC	0.618	0.665
Grading accuracy	75%	82%
Distinct entropy values reached	5	13
Median entropy, correct group	0.500	0.439
Median entropy, incorrect group	0.500	0.608

The exact median tie *does* break at $K = 15$ (Figure 2) – real evidence that $K = 5$ was not measuring nothing. But the improvement is modest, not dramatic: a pre-registered decision rule (AUROC ≥ 0.72 and the median tie breaking, to count as “sharper”; 0.55–0.72 as “real but

modest”; below 0.55 as “noise”) classifies this result as **confirmed_modest**. This pointed at a different question: is more sampling the right lever at all, or is the *digit itself* too coarse a readout of the model’s judgment?

4.3 Reasoning-text clustering: a stronger, less noisy signal (worked)

The grading prompt asks the model to write a **Reasoning** explanation before its **Error** digit. Every analysis so far discarded that explanation and kept only the digit. Clustering the reasoning text instead – via the same bidirectional NLI entailment approach that failed for perception (Attempt 2) – was tested against the already-collected $K = 5$ raw samples, at no additional GPU cost.

Metric	Digit-only	Reasoning-text
AUROC	0.618	0.702
Median entropy, correct group ($n = 75$)	0.500	0.950
Median entropy, incorrect group ($n = 25$)	0.500 (tied)	1.332

Given that the same NLI approach caused Attempt 2’s over-merging failure on perception, this result was not trusted without directly checking for the same bias. The check: do any two samples that *disagree on the parsed digit* (one says error, one says no error) ever get merged into the same reasoning-text cluster? They do – 55 times across the 61 items where samples disagreed on the digit. Inspecting a diverse sample of these merges (8 cases, different items) showed 7 of 8 were not opposed reasoning merged by mistake: *both* texts explicitly argued “there is an error,” just citing different specific details, while their own digits disagreed. Quantified across the full dataset: **60/494 samples (12%)** have digit= 0 while their own reasoning opens by flagging a mistake (“incorrect,” “error,” “mistake,” ...). The model is self-contradicting within single samples – writing a reasoning that argues for an error, then emitting the opposite digit anyway, plausibly decoding noise on the final token, independent of the fully-formed judgment already written. At $K = 5$, reasoning-text clustering is not an artifact of NLI bias; it is a readout that is not vulnerable to this specific noise source the way the digit alone is (Figure 3).

4.4 $K = 15$ GPU confirmation: the $K = 5$ result does not generalize (failed)

A GPU-accelerated confirmation at $K = 15$ (impractical on local CPU – a single item’s 105 pairwise NLI comparisons took 35s locally; GPU reduced the full 100-item pass to 58s) was run via a dedicated notebook (`pilot/02_reasoning_text_entropy.ipynb`). It reproduced the $K = 5$ result almost exactly (AUROC 0.7019 vs. the 0.702 found locally), then revealed the $K = 15$ result does not hold up:

Metric	$K = 5$	$K = 15$
AUROC (reasoning-text)	0.702	0.520
Items with a digit split	61/100	87/100
Cross-digit cluster merges	55	1187

0.520 is barely better than chance, and worse than the $K = 15$ digit-only result (0.665). The cross-digit merge count grew far faster than the $\sim 10\times$ increase in available pairs ($\binom{15}{2} = 105$ vs. $\binom{5}{2} = 10$) would predict – a $\sim 21\times$ increase in merges from a $\sim 10\times$ increase in comparisons. Re-clustering individual items locally to inspect actual cluster membership found the cause: **transitive**

cascade merging. The clustering method unions any two samples whose reasoning mutually entails, then merges transitively (union-find) – so a single mistaken pairwise link can silently absorb an entire subgroup into a cluster that does not actually agree. One item’s largest cluster contained 11 of 15 samples, mixing digit=1 samples arguing “*incorrectly multiplies by 5 instead of 9/5*” with digit=0 samples arguing “*does not contain any apparent...errors*” – genuinely opposed conclusions, merged together. This is not universal: a second inspected item’s largest cluster (7/15 samples) was entirely digit=0 with consistent reasoning, correctly merged. The failure is concentrated in a subset of items, but at $K = 15$ that subset is large enough to collapse the aggregate signal to near-random. More pairwise comparisons mean more chances for a single false-positive entailment call to cascade, and union-find has no mechanism to catch or bound that once it happens.

The $K = 5$ finding (AUROC 0.702) stands on its own – it was independently verified against real data, not merely computed once. But it does not generalize to $K = 15$ under the current clustering method, and should not be presented as validated at scale. A structural fix (e.g. requiring a minimum fraction of pairwise agreement within a cluster rather than any transitive chain, or a stricter/larger NLI model) would need to precede any further use of this technique at higher K .

5 Integrity Checks

Both entropy signals depend on the sampling and parsing pipeline actually working. Two checks validate that before trusting any entropy number.

Check	Result
Temperature-0 anchor, transcription (2 greedy draws/item, must be ≈ 0)	0/100 non-zero
Temperature-0 anchor, grading (2 greedy draws/item, must be ≈ 0)	0/100 non-zero
Transcription parse-failure rate (before marker-regex fix)	13.4% (67/500)
Transcription parse-failure rate (after marker-regex fix)	4.0% (20/500)
Grading parse-failure rate ($K = 5$)	1.0% (5/500)
Grading parse-failure rate ($K = 15$)	0.9% (14/1500)

The temperature-0 anchor passing at 0/100 confirms the sampling/parsing pipeline is behaving deterministically at temperature 0, as it must. The transcription parse-failure rate was itself a real bug (the extraction regex only matched the exact ****Answer:**** markdown the prompt requested, missing common real-world drift such as `\textbf{}` bold, plain text, and a full-width colon); widening the pattern recovered most of the missed samples.

6 Summary

Arm	Status	AUROC	Notes
Perception (final)	Working	0.788	Strong, visible separation
Perception (attempt 1)	Failed	–	83% pinned at max entropy
Perception (attempt 2)	Failed	–	Over-merged, 95% implausible
Reasoning, digit-only ($K = 5$)	Working	0.618	Real but weak, medians tied
Reasoning, digit-only ($K = 15$)	Working	0.665	Confirmed modest, near ceiling
Reasoning, text-clustering ($K = 5$)	Working	0.702	Best result at $K = 5$; validated
Reasoning, text-clustering ($K = 15$)	Failed	0.520	Cascade over-merging; does not generalize

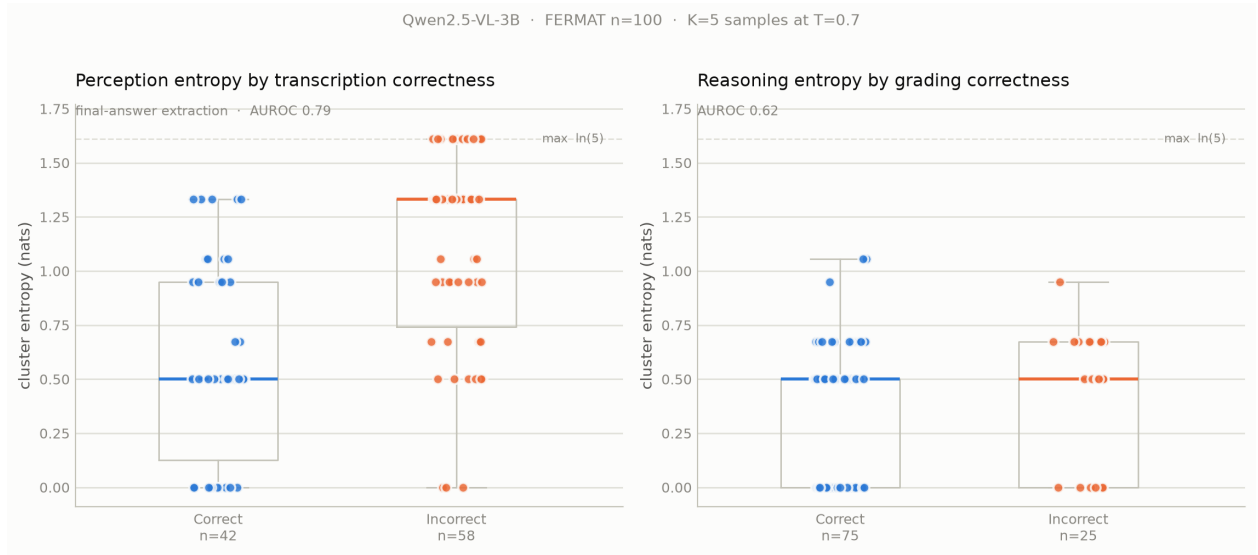


Figure 1: Final result for both arms, $n = 100$. Left: perception entropy by transcription correctness (AUROC 0.79). Right: reasoning entropy by grading correctness at $K = 5$ (AUROC 0.62).



Figure 2: Reasoning entropy before ($K = 5$) and after ($K = 15$) resampling, same 100 items. The exact median tie at $K = 5$ breaks at $K = 15$, but the overall separation remains modest.

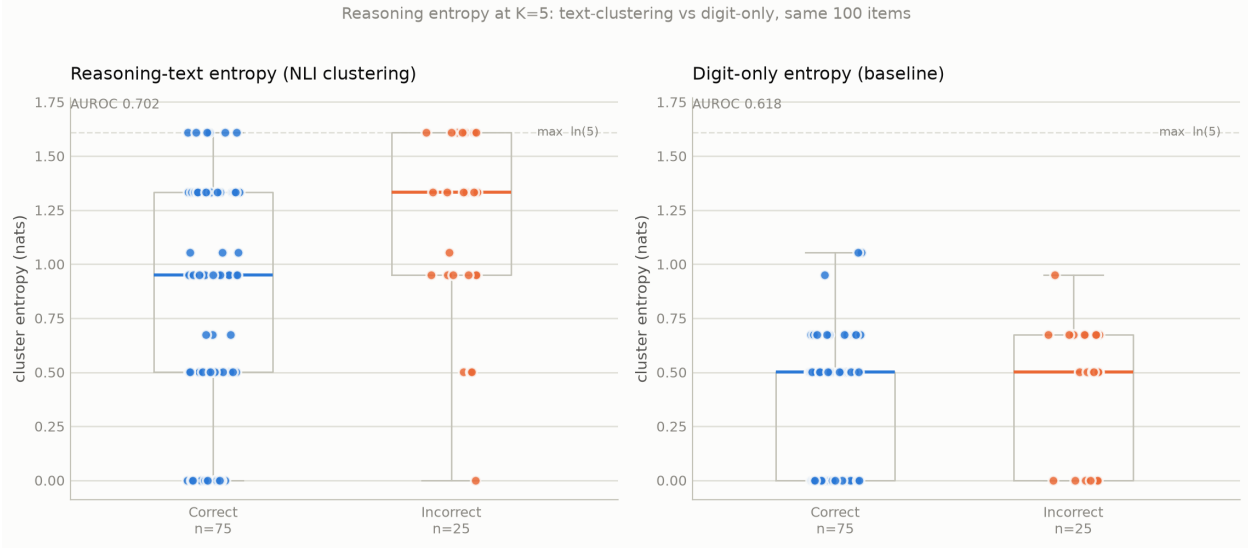


Figure 3: Reasoning entropy at $K = 5$: clustering the model’s reasoning text (left, AUROC 0.702) vs. the digit-only baseline (right, AUROC 0.618), same 100 items. The reasoning-text approach separates the incorrect group’s median well above the correct group’s; the digit-only approach ties them exactly.

7 Limitations

- **Single model, single run size.** All results are Qwen2.5-VL-3B on 100 items; no comparison to a larger model (7B) or a baseline uncertainty method.
- **No statistical inference.** No confidence intervals, no significance testing (DeLong’s test, bootstrap) – explicitly out of scope for this pass.
- **Perception correctness label is strict.** Transcription correctness requires the extracted final answer to match ground truth after canonicalization; some “incorrect” classifications are plausibly extraction-span mismatches (the model’s answer and the ground truth landing in different canonicalization tiers) rather than genuine disagreement, which would mean the true accuracy is somewhat higher than 42%. This biases the reported accuracy conservatively, not favorably.
- **No human double-grading.** Ground-truth labels are taken as given from the dataset.
- **Reasoning entropy’s ceiling is model- and task-specific.** The binary-outcome information limit applies to *this* grading task design; a differently-posed prompt (e.g. eliciting a confidence score rather than a forced binary decision) was not tested.
- **Reasoning-text clustering is confirmed to work at $K = 5$ only, and confirmed to fail at $K = 15$.** Not merely unconfirmed at scale – tested and found to collapse to near-random (AUROC 0.520) due to transitive cascade over-merging (§4.4). The underlying cross-encoder is also a small, general-purpose NLI model rather than one tuned for mathematical reasoning text, which may independently limit how far this approach can go even after the cascade issue is fixed.

8 Recommended Next Steps

1. **Fix the clustering algorithm’s cascade fragility before trusting reasoning-text clustering at any $K > 5$.** Replace or constrain the union-find transitive merge – e.g. require a minimum fraction of pairwise agreement within a cluster (a clique-style criterion) rather than any single chain of pairwise links, so one false-positive entailment call cannot silently absorb a whole subgroup. Re-run the $K = 15$ confirmation after the fix.
2. **Token-level confidence as an alternative reasoning signal.** Instead of inferring uncertainty from repeated discrete sampling, use the model’s own next-token probability for the 0/1 decision at generation time. A richer, more direct signal, immune to the clustering-cascade problem entirely since it does not require sample-to-sample comparison – but requires new sampling code and a new run to validate.
3. **Scale to a stronger model (7B) or the full dataset** only after resolving the above – the reasoning arm’s best *currently trustworthy* result is 0.702 at $K = 5$ specifically, not a number to scale up naively.